

Name: _____

Date: _____

Class: _____

MyDesign Portfolio

Element D Initial Template

Guidance before filling out the template on the next page:

Because you will iterate on your design multiple times, you will need to create multiple copies of these sections. Think of Element D as a diary that you will date and keep adding to throughout your work until your final iteration. It's critical that iteration shines through in this Element.

You should begin with many design ideas that reflect the individual ideas of your group members and then the group ideas. Number these ideas and include a date, description, drawing, and details for each.

After you have created an initial set of ideas, document your decision making on which design to include, ideally with a [decision matrix](#).

Create a plan of work, ideally using a [Gantt chart](#) that may be housed in this document or in a different day-to-day document.

Generate a design blueprint with enough details that someone else not familiar with your project could also build your design. Think of this as your first sketch.

You will then proceed forward to build and test this design, returning to Element D afterwards when you are ready to create more design ideas. Fill out each of these steps again.

Element D: Design Concept Generation, Analysis, and Selection

Design Idea Generation (D1)

This section focuses on the process of generating solutions. The initial rounds should include every idea you come up with but the subsequent rounds of ideation should lead to viable options. Each idea should include not only words, but also a picture. Use as many copies of this sub-template as needed.

Name: _____

Date: _____

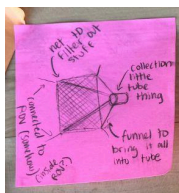
Class: _____

Design Idea #1: Dragging Plankton Net

Date: Feb 2

Description: This plankton net would be dragged around by our rover, dangling from the back of it. The net would funnel microplastics into a small collection bottle.

Drawing that includes distinguishing information such as scale, dimensions, and materials:

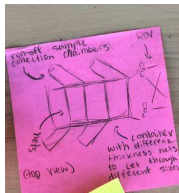


Design Idea #2: Multi-compartment

Date: Feb 2

Description: The 4 sieves would filter out different sizes of microplastics. We are unsure how or where this would mount to our ROV.

Drawing that includes distinguishing information such as scale, dimensions, and materials:



Design Idea #3 Interior Plankton Net

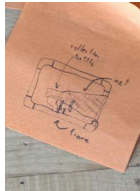
Date: Feb 2

Description: This plankton net would have its opening at the front of the ROV, looking a little like a mouth. The net would funnel microplastics into a small collection bottle. The benefit of having the net inside the ROV is that it is easier to maneuver the ROV this way. Drawing that includes distinguishing information such as scale, dimensions, and materials:

Name: _____

Date: _____

Class: _____

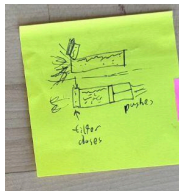


Design Idea #4 Filter Pump

Date: Feb 2

Description: would inhale contaminated water and push out the clean water and filter out the contaminants

Drawing that includes distinguishing information such as scale, dimensions, and materials: It would be a quarter-size of the ROV, and the piston would run on electricity. A filter would be at the end. A cap would



Design Idea #5 Venus flytrap filter

Date: Feb 2

Description: Two flaps will shut closed, trapping both water and microplastics.

Drawing that includes distinguishing information such as scale, dimensions, and materials:



Design Idea Comparison and Selection (D1, D2)

When you've finished generating ideas, you must compare your design options according to all your design requirements. Incorporate the priority of those requirements when comparing design options. A decision matrix (perhaps a Pugh Matrix) is encouraged since it allows for systematic, weighted comparisons. If using weights in a decision matrix, be sure



Name: _____

Date: _____

Class: _____

to explain and justify the weighting based on your design requirements. You may use the weighted design matrix template below, inserting your criteria and design options.

<https://docs.google.com/spreadsheets/d/10chf42psR-n9ONJrkXad0HPzQZ9hamvk/edit?usp=sharing&ouid=116396519872887702898&rtpof=true&sd=true>

Here is a link to our spreadsheet with each of our ratings of the criteria against each option for each teammate:

<https://docs.google.com/spreadsheets/d/10chf42psR-n9ONJrkXad0HPzQZ9hamvk/edit?usp=sharing&ouid=102487681563498430023&rtpof=true&sd=true>

For the weight of each criterion, we, as a group, went through each one, designating a weight percentage based on how much we valued each criterion. We gave each a percentage out of 100, then multiplied the rating out of 5 each member gave by that percentage to get the weighted value. We added every value for each idea for each member, then saw which one had the highest value for each person, and the one most chosen won.

Design Blueprint (D4)

Describe all of your design's details such that someone unfamiliar with your project could build your prototype. Include sketches, dimensions, CAD (if taught in your class), manufacturing methods, material choices, supply lists, and costs.

Name: _____

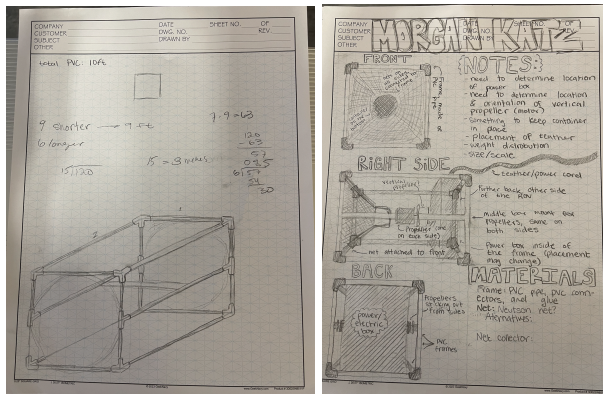
Date: _____

Class: _____

Design Blueprint #1

Date: Feb 9, 2026

Sketch:



Dimensions: 7 in x 7 in PVC square connected by 6 10 in PVC rods, one in each corner, and two through the middle of two opposite sides (doesn't matter which two, just have to be opposite of one another).

Materials: PVC pipe, pool noodles, propellers, nylon netting, a bottle, and wiring.

Manufacturing Methods:

Step 1:

Step 2:

Step 3:

Supply List and Costs:

Item	Cost each	# needed	Total
Nylon fabric (1 yard)	6.99	2	\$13.98
Total Cost			\$13.98

Name: _____

Date: _____

Class: _____

Plan of Action (D3)

A Gantt chart is encouraged to lay out the primary steps expected on a visual timeline. Larger tasks from the Gantt chart should be broken down further into detailed sub-tasks, which may be housed in a different day-to-day document. Discrete tasks should include enough detail such that others could replicate the efforts, such as procedures, materials, tools, status, etc.

<https://docs.google.com/spreadsheets/d/1OtSV5SzcPzgvyDo2T7SEsLOehhagz|cZih6ck|8DBPc/edit?usp=sharing>